



Potential Attacks of DeepFake on eKYC Systems and Remedy for eKYC with DeepFake Detection Using Two-Stream Network of Facial Appearance and Motion Features

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Abstract

eKYC, or electronic Know Your Customer, is an electronic or online customer identification allowing banks to completely identify customers online, simplifying paperwork or biometrics verification without having to meet face-to-face at the transaction office like traditional KYC. This new promising trend is becoming widely used in banks and various online services. Thus, eKYC can be a potential target for biometrics spoofing, including possible attacks by deepfake media, such as face swap or manipulation. In this paper, we introduce a potential attack of deepfake photos or videos on eKYC by swapping and manipulating faces between source and target faces. We then propose a novel method to detect fake videos of human faces using a two-stream network. We intend to utilize both the visual appearance and motion of a face in a sequence of pictures. We also propose to encode motion in an image sequence as an average photo which is called a motion-encoded image. Experiments on FaceForensics⁺⁺ show that using only a small number of frames in a video, our method can provide high accuracy in both frame-level and video-level prediction. With only 10 frames, our solution achieves the accuracy of 0.9326 and 0.9357 for the frame and video-level accuracy, respectively. If we use up to 50 frames per video, the accuracy of our method increases to 0.9371 and 0.9486 at the frame and video level. Finally, we propose to augment the security of current eKYC systems with deepfake detection. Our solution for deepfake detection can be a promising method to provide extra protection for eKYC solutions with face registration and authentication.

Keywords eKYC · Faceswap · Two-stream network · Deepfake detection

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Introduction

Electronic Know Your Customer (eKYC) is a form of electronic customer identification. This advancement in technology helps us simplify the procedures and paperwork, creating convenience for customers. Currently, banks in many countries, with the approval of competent regulatory authorities, have been transitioning to electronic customer identification. With eKYC, we can perform identity verification procedures right on the bank's application. Online identification technology will help banks simplify administrative procedures by paper. Besides, customers will not need to go directly to the bank counter to create an account, saving plenty of time.

In eKYC, it is essential to link a digital identity, e.g. an online account of an e-wallet, with the true identity of an individual. In the registration process for digital identity,

apart from the legal document, such as citizen identity (ID) card, passport, or driving license, the user is usually required to prove his or her true identity via face registration. This task is to prevent the wrong association between digital and real identities. As face registration or authentication can be tricked by photos or recorded video clips of the true user's face, the liveness of a face should be considered and integrated into such security processes [25].

Using human faces in eKYC systems is a new promising trend worldwide to provide users with a convenient way to interact with financial systems and services. Different strategies for challenging users are employed to protect the security of eKYC systems, such as blinking motion, multi-direction face motion, or read-aloud displayed OTP or PINs. However, videos generated by deepfake can become a potential threat for eKYC systems as an attacker can fool the system with computer-generated videos, even to respond to a challenge.

Without the existence of a real true person to register a digital identity, it would be possible to synthesize a fake face video of that person to bind that identity to an online account illegally [25]. Even if the eKYC solution employs the challenge that requests a user to perform some sequence of facial actions, e.g. smile, eye blink, turn left, look up, fake face video of a target person can also be synthesized with insignificant delay under the control of a source person.

In Ref. [14], we first introduce our ideas on the potential threat of deepfake images and videos for face-dependent eKYC systems and some of our preliminary results on solutions to detect deepfake photos and videos. In this paper, we extend our previous work [14] and develop our enhanced solutions for deepfake detection that can be applied to eKYC systems. Our main contributions are as follows:

- First, we propose the potential attack scenarios of deepfake on eKYC systems that use human faces for registration and authentication. We also study the current utilization of faces in eKYC systems of common banks in Vietnam to raise awareness on the potential threats to such systems.
- Second, we propose a mechanism to integrate deepfake detection in real time to protect eKYC systems from deepfake attacks. We focus our effort on simplifying the solutions to save time and computing costs for our solution. We currently use medium-sized backbone networks with about 20 million parameters (such as EfficientNetb4). We also consider the trade-off between the number of frames for evaluation with the overall accuracy in checking a real or fake face video.
- Third, we study different approaches and develop efficient methods for deepfake detection. In the preliminary results [14], we use Capsule Network [36] to enhance the capability of the network to learn features for real

and fake image discrimination. For our enhanced solution for deepfake video detection presented in this paper, we propose to encode motion in a sequence of frames into a single motion-encoded frame, and a two-stream network for deepfake detection. We intend to leverage both face appearance and motion information to check for potential issues in a fake video of a human face.

In our proposed solution for deepfake detection, we follow the common trend of utilizing multiple sources of information in two-stream network approach [9, 18, 19, 45, 47]. However, we develop our own design for the two processing streams in the network. The first stream in our network is to analyze the facial data in RGB format, while the second stream focuses on processing the encoded motion in consecutive frames. In this way, our intention is to combine both visual and motion data in a video sequence to identify real or fake face video.

The content of this paper is organized as follows. In the section “[Related Work](#)”, we briefly review the overview of eKYC and biometric factors, and then, we present the related work on deepfake and deepfake detection. In the section “[Deepfake Attack on eKYC Systems](#)”, we introduce a potential threat to eKYC systems using face swapping and manipulation to generate fake face video in real time to trick the registration and authentication processes of eKYC. We then propose in the section “[Deepfake Detection to Protect eKYC Systems](#)” our solution to enhance eKYC solutions with Deepfake detection techniques. The experiments of our Deepfake detection methods are discussed in the section “[Experiments](#)”. Finally, we conclude and present open problems for future work in the section “[Conclusion](#)”.

Related Work

Know Your Customer

Depending on the progress of the digitization process and the level of potential risks, some or all of the intermediary steps involved in the KYC process may be automated. eKYC, or electronic KYC is a concept of compliance with electronic KYC guidelines [27]. This form of KYC has become a de facto standard today. eKYC provides for financial organizations with not only customers' social information but also initial information about the customer's biometrics [40].

For a regular eKYC system, citizen portrait identification or any similar static 2D recognition method has the same weakness as written signature, in the sense that it is static and easily forgeable. One metric often used in judgment is the perceived “liveness” of the content [25]. Ideally, this metric should be zero for static images, and the opposite for a face captured live [17].

The two prominent methods for identification and authentication being currently adopted are Face ID and Video ID. These methods employ new and emerging technologies such as machine learning to complete the task.

Face identification data acquisition techniques capture the human face at multiple angles, giving both photometric (colors) and geometric (surface) information [12]. Video identification is a step up from face identification, requiring the user to perform an action and the result is a video [25]. This action commonly involves the user saying something then pick up and hold the identity card in front of the camera. This way, even more information can be analyzed from the content, such as gait, body identification, and proposed identity card. Video identification has the benefit of being very natural to the user while giving more information at the same time.

However, recently popular techniques such as Deepfake [2], which is currently state-of-the-art in the subject of face manipulation and face swapping, can generate realistic-looking faces that even confuse humans [8]. Deepfake requires visual images of a person to generate (“target person”), and the movements of the controlling person. It then proceeds to generate the target person’s face on the controlling person. This ability poses a threat to all existing face-based visual identification and authentication systems, as well as real-time face identification challenges.

Using faces for eKYC systems in banks in Vietnam: From our study, more and more banks in Vietnam implemented eKYC to provide convenient services to customers. From what we have seen, there are several authentication methods on various levels of “authenticity.” The following list is not complete, but shows an overview of the methods being used by common banks in Vietnam as of May 2022:

- Frontal view: VCB, BIDV, OCB, SHB
- 4-direction view: VietinBank
- Multi-direction face motion: VCB, ABBANK, BIDV, MBBANK, TECHCOMBANK, TPBANK
- Rightward face motion: VietCapital
- Blinking motion: VCB, ABBANK
- Smile: VCB
- Read aloud displayed OTP or PIN numbers: MBBANK.

These results show that front images are commonly used in most eKYC methods, while 4-direction facial images are used in the eKYC method of only one bank (VietinBank). Facial movement is the most utilized source of authentic information in the eKYC process employed by most banks. Some facial expressions, like blinking and smiling, are also used as a source of authenticity information. Notably,

the eKYC process of MBBANK also makes use of multi-modal interaction with visual and auditory inputs.

Deepfake Generation

Deepfake [26] refers to synthetic media, e.g. photos, audio, or video clips, using advanced algorithms in machine learning and artificial intelligence to manipulate or generate visual and audio content. Face swapping [31] or manipulation [34] is among the common and well-known demonstrations of Deepfake techniques [26]. We can easily notice the widespread use of face swap or manipulation in with social enthusiasm, from new commercialized applications for iPhones and Android smartphones, e.g. Reflect, Reface, to various YouTube channels with video clips synthesized with DeepFaceLab utility, etc.

Fake videos can be generated/synthesized by different methods with rapid advancement to reach photo-realistic level [26], e.g. FaceSwap [1], Face2Face [38], DeepFakes [2], and NeuralTextures [39]. DeepFaceLab [32] provides an easy-to-use open source framework that anyone (amateur) can use to make deepfake videos.

The typical pipeline for swapping faces from a source image to a target image usually consists of the following steps. First, the faces of two persons are extracted from both source and target images. Some methods [1, 2, 38] also require face normalization, facial landmark detection, and pose estimation. Finally, the face segment in the source image is transferred and blended into the corresponding segment in the target image. Deepfake generation can not only transfer facial features from a source to a target face but also head pose or facial movement [20] or expression [38].

In practice, some methods require training (finetune) the network to be well adapted to a pair of source and target persons. This approach is appropriate for creating high-quality Deepfake media. Using a pre-trained network for generic faceswap can be done, but the results may not be as good as the finetuned one. However, some recent methods no longer require training with new faces [31] like in early techniques [3]. Refinement with a few-shot or one-shot technique can also be applied for quality enhancement [43, 49]. For example, Zhu et al. propose hierarchical representation face encoder (HieRFE) to extend latent space to enhance the details in face [49], and face transfer module (FTM) to transfer identity from source to target without losing features.

Deepfake Detection

Several datasets have been released to evaluate different Deepfake detection methods. FaceForensics⁺⁺ [34] and DeepFake Detection Challenge Dataset [15] are among the common datasets for facial forgery detection.

To detect fake images, researchers usually focus on artifacts resulting from face tampering, such as from face swapping [48], face morphing [33, 38], splicing attacks [7], or Deepfake methods [21].

Most of the current methods for face manipulation detection in videos are CNN-based per-frame classification. Besides, inconsistency in facial movement, including eye blinking [22], head pose movement [41], or facial expressions [6], can also be exploited for the detection of fake videos. Several methods use CNN, 3D CNN [13], or RNN [35] to detect such inconsistency and exploit temporal dependencies in Deepfake video.

Attention masks can be used in various approaches for deepfake detection. For example, Wang et al. propose to find then delete attention regions in face photos to create new enhanced data for consideration. In this way, deepfake detector can focus on larger area, not only focusing on attention regions, and thus, the method can find possible artifacts in non-significant area in photo. However, as a complement approach, in Multi-Attentional Deepfake [44], Zhao et al. propose to process discriminative local regions with multiple attention maps, and enhance texture features from shallow layers to capture more noticeable artifacts.

For deepfake videos, the use of spoken words of the person through audio in a video is helpful information to detect fake videos. In LipsForensics [16], Haliassos et al. propose to use the area containing mouth to train lip-reading characteristics corresponding to each word. This feature is then used to judge whether the video is real or fake. This information-based method focuses only on mouth area of a face when the subject is speaking. Spoken content information can focus on a few specific phonemes that correspond to visemes (mouth shape) [4]. Zheng et al. take advantage of temporal coherence to detect fake faces in video [46].

Distinct facial or movement features generated by deepfakes are focused to classify whether a given frame or video was generated by deepfake or not. The following approaches provide us with the overview, yet not complete, of different artifacts or clues for deepfake detection in recent works.

Artifacts: In Ref. [24], Li et al. focus on the resolution limit of deepfake tools. With some low resolutions, the image needs to be scaled up. However, in the process of scaling up, depending on the interpolation method used, there are artifacts that are characteristic of the used interpolation method (e.g., bilinear, bicubic or lanczos). Because the information contained in some details is ultimately lost from the interpolation process, this void of information created by these transformations is manifested as unnatural regions, most noticeably along the edges where the synthesized face blends with the original face.

Blinking inconsistency: A normal person blinks every 2–10 s, and each blink lasts 0.1–0.4 s. In Ref. [23], Li et al. propose a method that scans through each frame in the video and find the position of the eye. Then, another deep neural

network is used to evaluate whether the eye detected in the frame is closed or open, based on various metrics, such as appearance, geometric features, or motion. This helps building a record of the state of the eyes with regards to frame time, with which both the duration and frequency of the blinking can be inferred out.

Difference between the original face and synthesized head: Contemporary deepfake techniques make use of face detection, selection, and synthesis pipelines. In Ref. [42], Yang et al. observe that the process of compositing the synthesized image onto the original face creates unnatural errors when applying 3d pose estimation. The synthesis step transforms a known image of a trained replacement face using the transformation matrix first extracted from the detection process. However, since the trained dataset does not always contain the replacement face with the desired pose, the selection process has to choose the one that most closely matched the pose in the original frame. Because of this, when 3D pose estimation is performed on the resultant video, errors can be observed.

Discrepancy between auditory and visual modal: Using videos has an advantage that information not only presented visually, but also exists auditorily. Chugh et al. in Ref. [11] and Agarwal et al. in Ref. [5] have observed that a change in speech or image leads to a breakdown of similarities between two modes of communication, such as mouth-to-mouth matching, unnatural facial expressions, and lips. In addition, the “dynamics” of seen mouthpieces, which do not match the sounds spoken, are also a source of information for evaluating deepfakes.

Consistency between spoken word and appearance: For deepfake videos, the use of spoken words of the person through audio in a video is helpful information to detect fake videos. In LipsForensics [16], Haliassos et al. propose to use the area containing mouth to train lip-reading characteristics corresponding to each word. This feature is then used to judge whether the video is real or fake. This information-based method focuses only on mouth area of a face when the subject is speaking. Spoken content information can focus on a few specific phonemes that correspond to visemes (mouth shape) [4].

Temporal coherence in video or 3D components: Zheng et al. take advantage of temporal coherent to detect fake faces in video [46]. Face forgery can be detected with 3D information. Zhu et al. propose to exploit potential extra information from 3D shape, such as common texture, identity texture, ambient light, and direct light.

Two-Stream Network for Deepfake Detection

Two-stream network is among the promising scheme for developing deepfake detection. Instead of using only one

processing branch with one main input, such as a face image, we can analyze and combine features from multiple information sources in two parallel branches of the network. As more information can be exploited, the output results are expected to be improved.

Zhou et al. propose a two-stream network to detect tampered faces in an image [47]. In the proposed two-stream network, the first stream learns and detects tampering artifacts in a face region. In the second stream, a patch-based triplet network is trained to exploit steganalysis features in different regions of the input image to detect tampered face with longer distance than other patches in the same image. It means that the Zhou et al. utilize not only face regions but also non-face ones in the background to identify tampered faces.

In the two-stream network proposed by Hu et al., the authors take advantage of the face reflectance images (MSR images) to exploit the illumination-invariant and high-frequency information for face spoofing detection [9]. Thus, together with the regular RGB image processing in one branch of the two-stream network, the authors add one more branch to handle the MSR image.

In Ref. [9], Hu et al. propose a two-stream method for deepfake video detection using frame inference. The first stream learn the face representation of a sequence of frames, and predict the future frames using auto-regressive model. The objective of the first stream is to predict a sequence of latent representations for a future sequence in the same video. The second stream of the method encode the real future sequence of frames. Based on the correlation of the two sequences of latent representations of the predicted and real future frame sequences, the network can classify a video to be real or fake.

Instead of using convolutional neural networks, Capsule network can also be adopted in the two-stream approach, as in Ref. [19]. Joseph et al. propose two branches, each of which uses a Capsule network, working in parallel to process RGB and Error Level Analysis images. This method is appropriate for detecting deepfake images and does not exploit inter-frame relationships.

In Ref. [45], Zhao et al. propose to use a two-stream convolutional neural network for deepfake video detection. The first stream detects semantic anomalies, such as face contour, detail missing, etc., while the second stream captures tampering traces in local noises.

Similar to existing work having two main branches in the processing network, we also propose a two-stream network for deepfake detection, as presented in the section “Two-Stream Network for Deepfake Detectionsec:TwospsStreamspNetworkspForspsDeepfakespsDetection”. The main difference in our proposed network is that we propose to utilize both an RGB image of a face and the

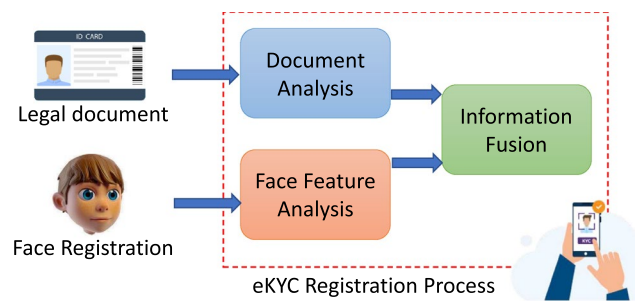


Fig. 1 Overview of a regular registration process in eKYC system

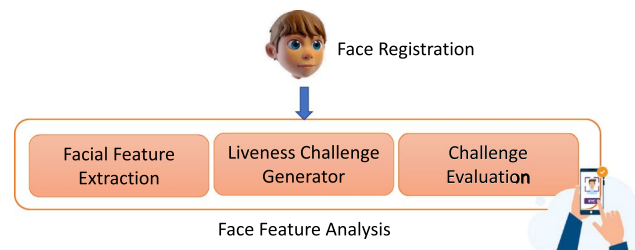


Fig. 2 Regular face feature analysis component

motion encoded from multiple frames in a video to identify potential tampered frame.

Deepfake Attack on eKYC Systems

Regular Registration Process for eKYC

We consider a typical registration process for an eKYC system, as illustrated in Fig. 1. There are two main inputs for this process including legal documents and biometric information, which are usually facial features. Document Analysis component extracts information about the person registering an online identity, e.g. full name, date of birth, address, citizenship, etc.; while face feature analysis extracts facial features for account registration. Information Fusion component provides extra necessary steps to link information between the two analysis components. For example, face verification may be employed in the registration process of some eKYC systems to confirm that the live face of the person who is registering the account actually matches the face in the legal document.

In Fig. 2, we present the main modules for a regular face feature analysis component. Apart from the facial feature extraction to register a user’s face, it is crucial to verify the liveness of that face during the registration process. The live challenge generator produces a sequence of requests R for specific facial gestures that a user should perform to confirm the liveness. Each request in the challenge R is

selected from a known challenge face gesture set S , including smile, left/right eye blink, right eye blink, open mouth, turn head left/right/up/down, etc. The number and order of requests in a challenge R can be pre-defined or generated adaptively depending on the confidence evaluation and environment of the user.

Attack Registration Process with Deepfake

In this section, we present two scenarios for an adversary to trick the face registration process of an eKYC system. In scenario 1, as shown in Fig. 3, the registration process requires a customer to capture and submit a number of photos of his/her face (see the left part), which allows for 3D reconstruction of the customer's face.

The essence of this challenge is to create multiple differentiators between real and live data versus generated and/or collected data to present as few difficulties to legit customers as possible while being very hard to complete for an adversary. These differentiators could be, for example, the perceived liveness index of the photos, the difference between the requested face angle versus the received face angle, the consistency among the photos, or the amount of time to produce an answer. A legit customer has no difficulties accomplishing this challenge by taking pictures of his/her face in a reasonably short amount of time. This was not the case for adversarial methods. Methods that are based on

a large collection of a victim's data rely on the chance that there exists a sequence of face photos, with minimal difference in requested angles presented in the challenge.

Nevertheless, if the incentive is high enough, a malicious party could exploit the 3D reconstruction process by manually manipulating the collected photos to adapt to the requested angle. This method satisfies the differentiator established in the difference between the requested angle and the received face angle. However, it is likely time-consuming, and thus is differentiated by the time differentiator and failing the challenge.

Figure 4 presents another scenario where the registration process requires new customers to film a video proof. In the video, a user is requested to perform some actions, such as holding an ID card and tilt his/her head. This method is commonly used by not only banks but also exchanges for customers that are willing to trade in high volume.

Video-based proofs further enhance the consistency and the amount of data available to analyze, while making minimal changes to the customer experience. Owing to the advance of ubiquitous mobile technology developments, the process of taking video proof differs in little with taking multiple photos. Similar to taking multiple photos, this approach generates multiple photos from the video, and 3D reconstruction of the customer's face is possible and can be performed.

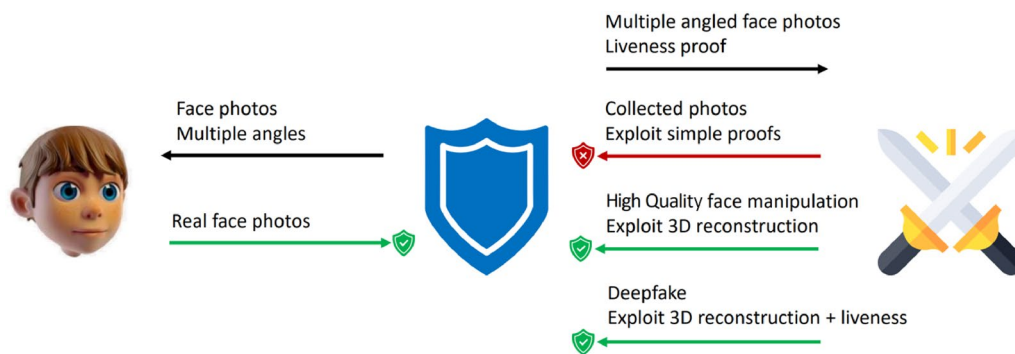


Fig. 3 Scenario 1: attack on face registration with a photo

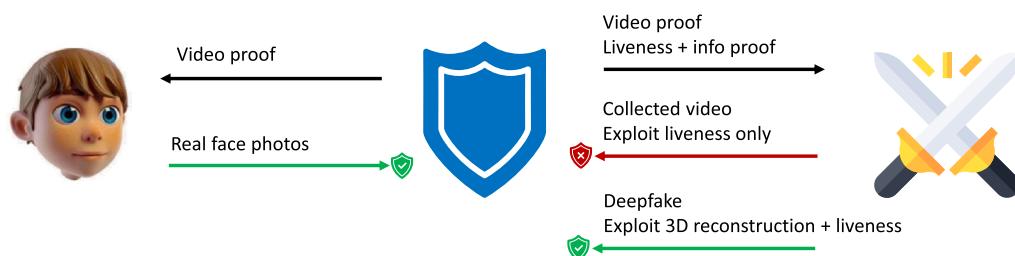


Fig. 4 Scenario 2: attack on face registration with a live video

Using video, various movements contained within can be easier analyzed for discrepancies. Likewise, any legit customer can complete this challenge with little difficulty, while it is unlikely that there exists a pre-recorded video that an adversary possesses that matches the requested head poses.

However, common to the two aforementioned methods, now exists a threat from Deepfake. For an adversary that uses a controlling person and possession of a target face, Deepfake has shown to be very flexible in generating faces on the controlling person using the target face. With a resourceful adversary party, the time required to perform face swapping can be reduced to real time, and thus render the time differentiator ineffective. The quality of targets that Deepfake can generate is good enough that it even allows 3D reconstruction to take place.

In Fig. 5, we propose a potential attack on the face registration process in eKYC system using Deepfake techniques. From face photos and/or video clips collected from a true person *B*, we can train a model to learn how to synthesize/swap face from *B* to a face motion video of an adversary *A*. This training phase can be time-consuming, but can be done offline.

During the face registration process in an eKYC application, an eKYC server generates a sequence of facial gesture challenge requests to check the liveness of a user. Instead of having the true person *B* to actually perform face gestures according to requests from eKYC server, an adversary *A* can impersonate *B* by capturing his/her face as a sequence of frames, which are then fed into a Deepfake system to synthesize a corresponding video with the target face of *B*. This sequence of frames can be generated with low latency in response to the request from the eKYC server.

Our experiment can generate a Deepfake video with a length of 10 s in less than 5 s with a system of two NVIDIA 2080Ti GPUs. Currently, we aim to create high-quality fake video clips, so we use a desktop implementation. If we want

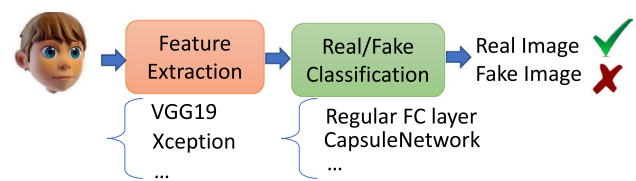


Fig. 6 Common framework for Deepfake detection

to further boost the Deepfake video generation, we may employ knowledge distillation technique to train a smaller student network for Deepfake face image synthesis from a larger teacher network. We also plan to utilize mobile apps for a Deepfake video generation in future work.

Deepfake Detection to Protect eKYC Systems

In this section, we first present the common framework for Deepfake detection, and our strategies to realize this framework with different components. Then, we propose to integrate deepfake detection module as an add-on component to enhance the security of an eKYC system with face registration and authentication.

Initial Methods for Deepfake Detection

In this section, we recall our initial methods for deepfake detection that we presented in Ref. [14]. From the experiments and results with these methods, we are encouraged to continue study and propose new method presented in the section “Two-Stream Network for Deepfake Detection”.

Deepfake detection has become an attractive problem for many researchers and practitioners worldwide. Therefore, more than 2000 teams already participated in the competition of deepfake detection challenge (2019–2020).

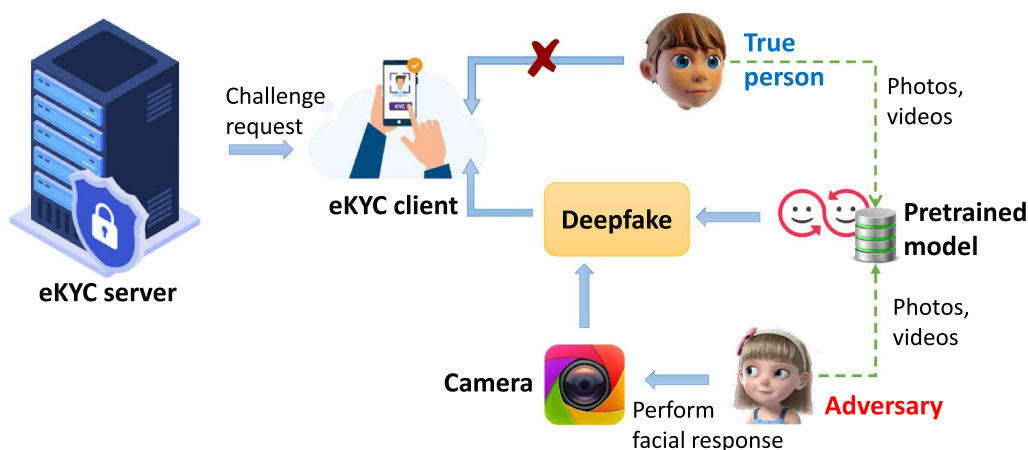


Fig. 5 Scenario 3: attack with Deepfake

Despite different techniques and strategies of implementation, we can find the common approach for current Deepfake detection methods, as illustrated in Fig. 6. There are two main modules in the framework. The first module extracts facial features, while the second one is for real/fake image classification. In our implementation, we employ different backbone for feature extraction, e.g. VGG19 [37], Xception [10], etc.. For classification, we consider the classical approach with fully connected layer(s) and also experiment the CapsuleNetwork [28, 36].

To train a Deepfake detection system with two main components, as shown in Fig. 6, we divide this task into two phases. In phase 1, we reuse pre-trained weights for the feature extraction module and freeze them, then only train the classification module. By this way, we expect to initialize the feature extraction module with good-enough weights, and focus on training the second module. In phase 2, we unfreeze the feature extraction module, and finetune the whole network.

In our first implementation, we use a VGG19-like network for feature extraction. As shown in Fig. 7, we consider up to three levels of details in a face; thus, we use the output after the 3rd MaxPooling layer of the VGG-like network as the face feature for classification. For a deeper analysis of face, we may use features after the 4th or 5th MaxPooling layer. We do not employ the feature flattening steps as we want to preserve much face information. In our future work, we also consider combining features output

from different levels of details, such as after each Max-Pooling layer, to utilize more information about the face.

To better learn how to classify between a real and fake face images, we employ CapsuleNetwork [36] for the classification task, as illustrated in Fig. 8. This approach can be considered as the implementation following the idea of Capsule-forensics, proposed by Nguyen et.al in Ref. [28], for detecting fake image and video. In the traditional approach for convolutional neural networks, features extracted are usually flattened, then processed through one or several fully connected layers, and finally evaluated by a softmax one. CapsuleNetwork utilizes the idea to mimic the brain's operation, i.e. the brain should employ a mechanism for "routing" low-level visual information to what it believes is the best capsule, a nested set of neural layers, for handling it. Dynamic routing, or routing by agreement, is considered to be superior to the current mechanism like max-pooling that routes based on the strongest feature detected in the lower layer.

Two-Stream Network for Deepfake Detection

In the section "Initial Methods for Deepfake Detection", we briefly recall the method [14] we used in deepfake detection challenge (2019–2020). With the promising results from this challenge, we continue to study and propose a new method presented in this section, namely two-stream network for deepfake detection. Our new method aims to

Fig. 7 Face feature extraction with VGG19 in Deepfake detection

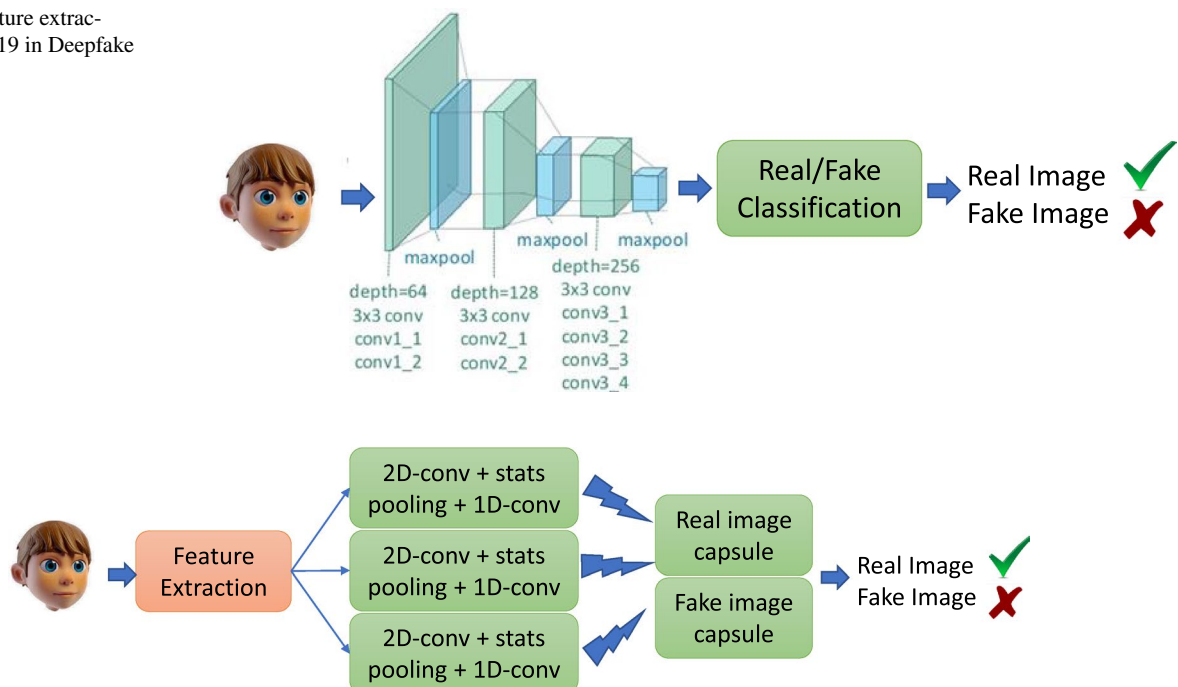


Fig. 8 Deepfake detection with CapsuleNetwork

utilize both appearance and motion information in the face region to verify if a face image in a video is real or fake. This new approach overcomes the limitation of our previous method [14] that exploits only the visual appearance feature of a face. Our new method is based on the observation that in a fake video, i.e. face manipulated video, the motion of each face region may contain *inconsistency* or *irregular patterns* across frames. For example, some parts may move faster or slower than their neighbors.

In the conventional approach (as described above), the face information is passed through the feature extraction module, and then, the extracted feature vector is fed to the real/fake image classifier. We propose the use of two inputs, one is the cropped face area in frame t , and the other receives motion information in the face area in the previous consecutive frames, i.e. frames from $t - \Delta t$ to t .

There are different ways to represent and extract features of visual motion in an image sequence. For example, we can use LSTM to encode frame-by-frame in the sequence sequentially. In this paper, we propose a simple way to encode motion from the sequence using a weighted averaging image [29, 30].

Figure 9 illustrates the process to generate motion-encoded image for an image sequence with $\Delta t + 1$ frames from frame $t - \Delta t$ to frame t . We detect the bounding box of a face in the current frame t and use that bounding box as the window to crop the previous Δt frames. In this way, all cropped regions are well aligned and we can add all cropped face regions to create a motion-encoded image. We can also use different weights for each previous frame. The resulting image seems to contain motion blur, which can be considered to represent motion information in the sequence.

In our proposed two-stream network for deepfake detection, we use two feature encoders to process a face image and a motion-encoded image, as shown in Fig. 10. These two feature encoders have the same network structure; for example, we can use EfficientNetB4 here. However, in the training process, as we unfreeze the whole network, the two feature encoders are finetuned, and finally, they have different weight values. We combine the face appearance and face motion features and feed this combined feature to the real/fake classification module to get the final result.

If the face has no significant movement, the motion-encoded image may not contribute much to the classification of the current frame as real or fake. However, if the face has

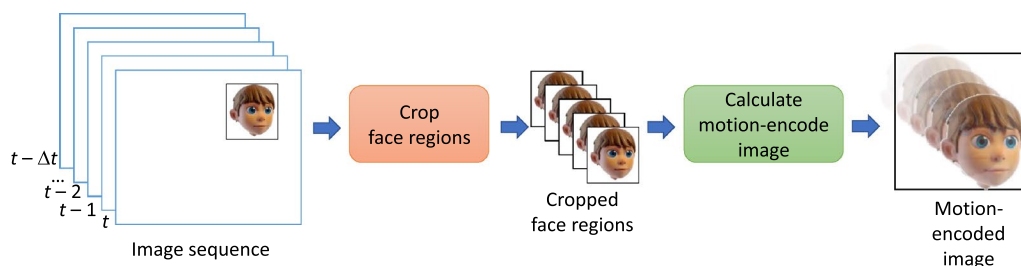


Fig. 9 Motion-encoded image in a face region for a sequence of images

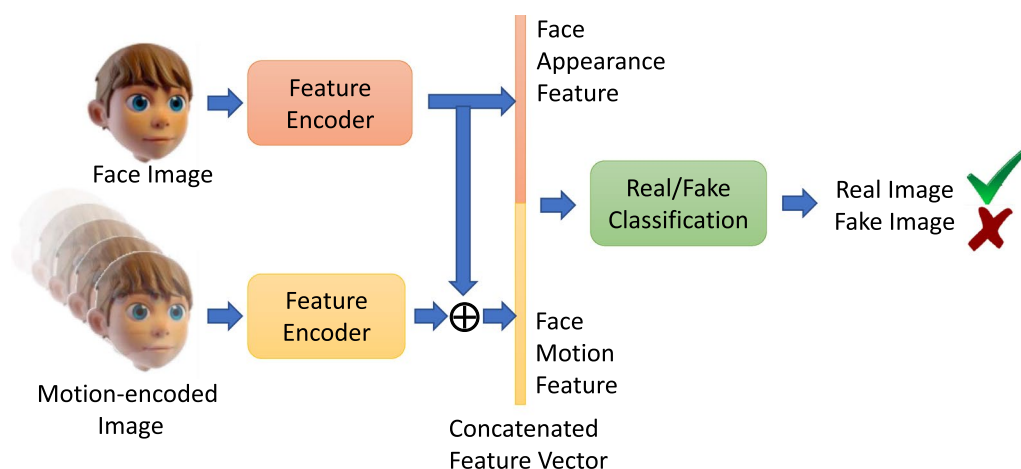


Fig. 10 Deepfake detection with two-stream network

noticeable partial or global movements, the motion information from the second branch provides valuable input for our model. Therefore, in the design of this network, we treat the outputs of the feature encoders in the two branches differently. In the first branch (for face image), we directly use the output of the feature encoder as the face appearance feature. In the second branch (for motion-encoded image), we combine (add) the outputs of the two branches to form the face motion feature. The face appearance and face motion features are concatenated into a single concatenated feature vector, which is then fed into the real/fake classification module.

From Frame-Level to Video-Level Accuracy

Using the two-stream network as proposed in the section “[Two-Stream Network for Deepfake Detection](#)”, we can classify a photo to be real or fake using that photo and its previous frames (for motion information). To process a video, we perform the frame-level classification multiple times with n_{Samples} frames and use the majority voting scheme to determine if that video is real or fake based on the majority label from the classification results of selected frames.

We determine the frame-level accuracy by counting the number of frames that can be classified correctly and dividing it by the total number of examined frames. We also define the video-level accuracy as the number of correctly classified videos over the total number of videos.

In the section “[Relationship between Frame-level and Video-level Accuracy](#)”, we analyze the relationship between frame-level and video-level accuracy.

Enhanced eKYC with Deepfake Detection

To strengthen the security for eKYC systems, particularly with face registration and authentication, we propose to integrate several components to validate face photos and photo sequences, as represented in Fig. 11. In practice, we have a sequence of face images returned from a mobile eKYC

client, responding to a face gesture request from eKYC server. The deepfake detection module is responsible for verifying if a face image or a face sequence is real or fake.

Using our proposed method with a two-stream network, presented in the section “[Two-Stream Network for Deepfake Detection](#)”, we can classify a single photo to be real or fake. We then use the majority voting scheme for deepfake video detection at video level. From the section “[From Frame-level to Video-level Accuracy](#)”, we can see that if our single-frame classifier provide a good-enough accuracy, yet not necessarily perfect score, we have high probability to correctly evaluate a video to be real or fake.

In our experiments, presented in the section “[Experiments](#)”, using $n_{\text{Samples}} = 10$ can provide a high reliability for frame-level prediction with the best accuracy of 0.9326, and the average accuracy of 0.9243 ± 0.0039 on FaceForensics++ dataset. Using this frame-level prediction, our method achieves the video accuracy of 0.9357, i.e. predict correctly 655/700 videos in the dataset. With the promising results on FaceForensics++ dataset, we will integrate our solutions to practical systems with careful finetune and optimization.

In our proposed solution to enhance eKYC, we also define face motion stability check. This module is responsible for checking the continuity and stability of visual data to trigger notification when there is significant sudden change in motion.

With the assistance of deepfake detection and face motion stability check, we propose to generate more liveness challenges, if necessary, in face feature analysis step. To provide the flexibility for our system to handle new face manipulation methods, we develop an add-on manager that can be used to plug-in more security functions.

Experiments

In Ref. [14], we present the results of our early stage method for deepfake detection on the DFDC dataset in the deepfake detection challenge (2019–2020). In our implementation, we use VGG19-like network [14] and Xception. We employ two techniques for classification, namely classical fully connected layers and CapsuleNetwork [36], following the original idea of Capsule-forensics [28]. In the public leaderboard, we achieve the Logloss of 0.40214. For the official leaderboard with the private test set, our method achieves a Logloss of 0.51880 with Bronze Medal, among the top 6% teams in the challenge. This encouraging initial result is the motivation to help us deeply study and analyze necessary techniques and other methods to propose our new solution in this paper. In our novel solution with a two-stream network, one input is a cropped human face photo and the other input is a motion-encoded image.

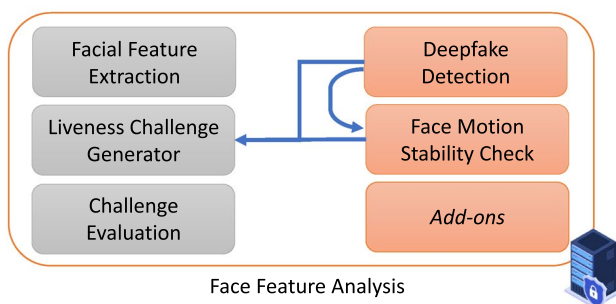


Fig. 11 Enhanced face feature analysis component with deepfake detection

In this section, we first present the dataset and how we use it in our experiments in the section “[Dataset](#)”. Then, in the section [Evaluation of Different Backbone on Deepfake Detection](#)”, we study the accuracy for deepfake detection with different backbone networks, so that we determine a prominent one for the next steps. We study the trade-off between number of examined frames in a photo sequence and the deepfake detection accuracy in the section “[Trade-off between Number of Examined Frames and Deepfake Detection Accuracy](#)”, and conclude that using only small number of frames can be sufficient to provide high accuracy for detecting fake photo/video, and thus suitable for eKYC systems. We also study the capability of deepfake detection on different specific manipulation techniques to generate deepfake photos or videos in the section “[Deepfake Detection with Each Deepfake Generation Method](#)”. Finally, we compare to significantly highlight the advantages of our two-stream network for deepfake detection over the common trend with a single input for networks in the section [Deepfake Detection with Each Deepfake Generation Method](#)”.

Dataset

In this paper, we use the FaceForensics⁺⁺ dataset [34] to conduct various experiments. In this dataset, there are four methods for face manipulations to generate fake videos, namely DeepFakes (DF), Face2Face (F2F), FaceSwap (FS), and NeuralTextures (NT). The dataset consists of 5000 videos, including 1000 original/pristine, i.e. real videos, and 1000 fake videos of each deepfake type. Among 1000 videos in each category, the number of videos for training, validation, and testing are 720, 140, and 140, respectively.

We denote $\mathcal{D}_0$ as the whole FaceForensics⁺⁺ dataset with all videos in four deepfake generation techniques and real/origin videos. We also define $\mathcal{D}_1$, $\mathcal{D}_2$, $\mathcal{D}_3$, and $\mathcal{D}_4$ to be the subset of FaceForensics⁺⁺ dataset, corresponding to DF, F2F, FS, and NT face manipulation techniques. Precisely, each subset $\mathcal{D}_x$ for $1 \leq x \leq 4$ contains real videos and fake ones only from a specific deepfake technique.

We aim to find solutions to detect deepfake videos in the context of protecting eKYC systems, where videos are often recorded on mobile devices. Therefore, we try to test and evaluate our solution in a situation that is close to reality, i.e. without requiring the video to be of truly perfect quality. In the FaceForensics⁺⁺ dataset, there are three quality levels: RAW, high quality (HQ or Compressed 23), and low quality (LQ or Compress 40). In fact, RAW data are not of perfect quality, because videos are downloaded initially from YouTube and are already compressed. It is obvious that using a higher quality video may help us achieve higher deepfake detection results. However, because we want to evaluate the deepfake detection method in a context that closely resembles the actual situation in eKYC systems, where users record videos from mobile devices, we decide to choose data at the HQ level in FaceForensics⁺⁺.

For each video, we select n_{Samples} first frames to evaluate the real/fake property. From the recommendation in the appendix of [34], n_{Samples} is 270 and 100 for a training and testing video, respectively. However, in this article, we intend to evaluate whether just taking fewer frames in a video for consideration has a big impact on the evaluation results or not. Suppose we only need to select a small set of frames, and the results are still high and meet security requirements. In that case, it enables us to deliver a more effective fake video detection solution for an eKYC system instead of considering most or all frames in a video.

In our experiments, for simplicity, we denote $\mathcal{D}_x[n_{\text{Samples}}]$ as a $\mathcal{D}_x$ dataset (for $0 \leq x \leq 4$) where only n_{Samples} frames of each video are used for training and testing. For example, with $\mathcal{D}_0[50]$, we consider all data in the FaceForensics⁺⁺ dataset, but we only use 50 frames of each video. With $\mathcal{D}_2[10]$, we use a sub-dataset consisting of only 1000 Face2Face and 1000 origin videos, and process only ten frames of each video.

Figure 12 illustrates the full dataset, denoted as $\mathcal{D}_0$, and its sub-datasets $\mathcal{D}_x$ for $1 \leq x \leq 4$ corresponding to

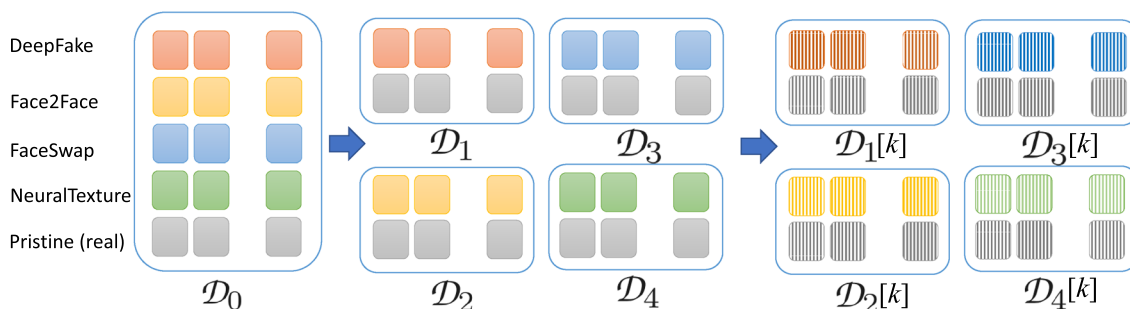


Fig. 12 Dataset and strategy to conduct experiments

four different face manipulation techniques. We extract $n_{\text{Samples}} = k$ frames from each video for training and testing.

In our experiments, we use two types of machines:

- Intel(R) Xeon(R) Silver 4210R CPU @ 2.40GHz, 192GB ram, RTX5000 16GB: for experiments on $\mathcal{D}_x$ for $1 \leq x \leq 4$.
- Intel(R) Xeon(R) W-2102 CPU @ 2.90GHz, 32GB ram, RTX A5000 24GB: for experiments on $\mathcal{D}_0$.

Evaluation of Different Backbone on Deepfake Detection

As described in the section “[Introduction](#)”, our objective is to develop an accurate but effective solution for detecting deepfake images and videos for eKYC system. It means that the network model should not be too heavy and the computational cost should not be too high. Therefore, we decide to consider medium-sized network models, about 20 million parameters. Specifically, we choose to test with Xception (22.9M parameters), EfficientNetB4 (19.5M parameters), DenseNet201 (20.2M parameters), and Resnet50 (25.6M parameters). In the future, we will continue to study smaller size models using the smaller available models, or train small network distilled from a larger successful model.

Table 1 shows the experimental results on $\mathcal{D}_0[10]$. We process all videos in FaceForensics⁺⁺, but only select $n_{\text{Samples}} = 10$ frames in each video to train or test. This allows us to have different videos generated by all four deepfake techniques, and save time to test and compare the accuracy of different backbone networks. From the experimental results, we decide to use EfficientNetB4 for experiments in the following sections.

Trade-Off Between Number of Examined Frames and Deepfake Detection Accuracy

The objective of this experiment is to verify if we need to process many frames in an image sequence to check whether a video is real or fake with high accuracy. Thus, we consider different values of n_{Samples} to train and to test videos. In our experiment, we choose $n_{\text{Samples}} = 10, 30$, and 50. These

Table 1 Deepfake detection with different backbone networks

Model	Parameters	Best accuracy	Average accuracy
Xception	22.9M	0.9184	0.8933 ± 0.0215
ResNet50	25.6M	0.9027	0.8667 ± 0.0350
DenseNet201	20.2M	0.9143	0.8982 ± 0.0108
EfficientNetB4	19.5M	0.9291	0.9169 ± 0.0109

Table 2 Deepfake detection with different number of sample frames in each video

Dataset	n_{Samples}	Best accuracy	Average accuracy
$\mathcal{D}_0[10]$	10 frames/video	0.9291	0.9169 ± 0.0109
$\mathcal{D}_0[30]$	30 frames/video	0.9412	0.9377 ± 0.0042
$\mathcal{D}_0[50]$	50 frames/video	0.9437	0.9403 ± 0.0029

values are smaller than the recommended number of frames for training (270) and testing (100) in Ref. [34].

Table 2 shows the experimental results on the whole dataset, i.e. $\mathcal{D}$, using EfficientNetB4 and different number of frames to be processed in each video. We consider $n_{\text{Samples}} = 10, 30$, and 50. It can be seen that if we increase the number of samples per video, the accuracy tends to increase, but not significantly. Thus, we can select a small number of frames in a video, or a live sequence of photos, to quickly verify if a video is real or fake with high accuracy. Therefore, in the next three subsections, we prefer using $n_{\text{Samples}} = 10$ in our experiment to other higher value of n_{Samples} . We also show in the section “[Two-Stream Network for Deepfake Detection](#)” that if we use a method with a *high but not necessarily perfect* accuracy to classify a real or fake photo, we can achieve very high accuracy to check a video using a majority voting scheme.

Deepfake Detection with Each Deepfake Generation Method

In the previous two sections, we conduct experiments on $\mathcal{D}_0$, i.e. the entire FaceForensics⁺⁺ dataset. In this section, we evaluate the discriminating ability of real and fake video for each method of generating fake videos, and also evaluate the ability to handle new cases, i.e. other methods of generating fake videos.

Table 3 shows the experimental results for deepfake detection on four different sub-datasets $\mathcal{D}_x$ for $1 \leq x \leq 4$,

Table 3 Deepfake detection with different deepfake generation methods

Dataset	Manipulation method	n_{Samples}	Max accuracy	Average accuracy
$\mathcal{D}_1[10]$	DeepFake	10	0.9657	0.9610 ± 0.0059
$\mathcal{D}_2[10]$	Face2Face	10	0.9675	0.9573 ± 0.0082
$\mathcal{D}_3[10]$	FaceSwap	10	0.9686	0.9599 ± 0.0059
$\mathcal{D}_4[10]$	NeuralTexture	10	0.8971	0.8801 ± 0.0121
$\mathcal{D}_1[50]$	DeepFake	50	0.9931	0.9893 ± 0.0026
$\mathcal{D}_2[50]$	Face2Face	50	0.9801	0.9752 ± 0.0041
$\mathcal{D}_3[50]$	FaceSwap	50	0.9933	0.9906 ± 0.0032
$\mathcal{D}_4[50]$	NeuralTexture	50	0.9149	0.9050 ± 0.0065

corresponding to the four manipulation techniques DeepFake, Face2Face, FaceSwap, and NeuralTexture. For each sub-dataset, we consider two configurations to process the $n_{\text{Samples}} = 10$ or 50 frames per video in training and testing.

There are eight cases in Table 3. For each case, from 1000 real (Pristine) and 1000 fake videos, we use 2×560 videos for training and 2×140 videos for testing. We report the accuracy of the best trained model that we can find, as well as the average accuracy (and standard deviation) of different models trained within the test case.

The results of all eight cases are pretty high: 6/8 cases have the maximum and average accuracy of 0.95 or higher. Among the four techniques for generating fake videos, only the NeuralTexture method is difficult to distinguish real/fake videos. However, the accuracy on NeuralTexture data is still promising (with the best accuracy of 0.8971 and 0.9149 on $\mathcal{D}_4[10]$ and $\mathcal{D}_4[50]$, respectively). This shows that deepfake detection can be done with high accuracy. Furthermore, the difference between the average and maximum values of accuracy is insignificant, along with a relatively small standard deviation, indicating that the method is relatively stable.

Two-Stream Network for Deepfake Detection

In Table 4, we illustrate the comparison of the accuracy at frame level between the single input method and two-stream approach. For the latter method, we apply two sampling strategies:

- **Early sampling**, denoted as Two-Stream_E: the first n_{Samples} of a video clip are selected for evaluation. This strategy aims to provide the early prediction for a video clip.
- **Uniform sampling**, denoted as Two-Stream_U: the n_{Samples} frames are selected uniformly in a video clip. This strat-

egy aims to collect frames with wide variation in a video clip.

As mentioned in the section “[Trade-off between Number of Examined Frames and Deepfake Detection Accuracy](#)”, with a small number of examined frames in a video or a live photo sequence, if we achieve high accuracy, yet not necessarily perfect, to classify a photo at frame level, we have a great potential to verify a video to be real or fake with very high accuracy. Therefore, we decide to conduct this experiment on $\mathcal{D}_0[10]$, i.e. only process 10 frames per video in training and testing. For the single input method, the best accuracy is 0.9291, and the average accuracy is 0.9169 ± 0.0109 . Using our proposed method with two-stream inputs, we can boost the best accuracy to 0.9326 and 0.9321 for two-stream_E and two-stream_U, respectively. The average score is also enhanced to 0.9243 ± 0.0039 and 0.9248 ± 0.0044 . Using more frames in a video, e.g. $n_{\text{Samples}} = 30, 50$ or even 100, we may further increase the success rate of classifying photo to be real or fake. In Table 4, we also provide our experimental results on $\mathcal{D}_0$ with $n_{\text{Samples}} = 50$.

From the experiments, we can see that the uniform sampling strategy tends to provide better accuracy than the early one, but the difference is insignificant. Therefore, the uniform method is more suitable to process recorded video clips to get higher accuracy, and the early strategy is of preference to provide high accuracy for early prediction on live data, such as in eKYC authentication processes.

Relationship Between Frame-Level and Video-Level Accuracy

In all previous experiments, we only use frame-level accuracy. We count all frames that can be predicted accurately to be real or fake, and then divide it with the number of processing frames. Using the majority voting scheme, as presented in the section “[From Frame-level to Video-level Accuracy](#)”, we can classify a video to be real or fake. This section studies the relationship between frame-level and video-level accuracy in deepfake video detection.

If we evaluate n_{Samples} frames in a video independently and get more than half of the votes to be correct, we can then classify the whole video accurately. For example, for a video in $\mathcal{D}_0[10]$, we need at least 6 frames predicted accurately. With a video in $\mathcal{D}_0[50]$, the minimum number of successful frame-level prediction is 26.

Let the variable X denote the number of successful predictions at frame level. We define $n_{\text{Threshold}} = k + 1$ for $n_{\text{Samples}} = 2k$ or $2k + 1$. We need $X \geq n_{\text{Threshold}}$ to ensure a correct video-level prediction using a majority voting scheme. For example, when we process $n_{\text{Samples}} = 10$ frames in a video, and the number of correctly classified frames X is 6 or more, the final decision for the video is correct. In

Table 4 Comparison between single input and two-stream approach on $\mathcal{D}_0[10]$

Method	Dataset	n_{Samples}	Best accuracy	Average accuracy
Single	$\mathcal{D}_0[10]$	10 frames/video	0.9291	0.9169 ± 0.0109
Two-stream _E	$\mathcal{D}_0[10]$	10 frames/video	0.9326	0.9243 ± 0.0039
Two-stream _U	$\mathcal{D}_0[10]$	10 frames/video	0.9321	0.9248 ± 0.0044
Single	$\mathcal{D}_0[50]$	50 frames/video	0.9437	0.9403 ± 0.0029
Two-stream _E	$\mathcal{D}_0[50]$	50 frames/video	0.9371	0.9343 ± 0.0022
Two-stream _U	$\mathcal{D}_0[50]$	50 frames/video	0.9449	0.9386 ± 0.0041

another example with $n_{\text{Samples}} = 50$ frames to be processed in a video, the minimum number of correct votes should be $n_{\text{Threshold}} = 26$.

Let p be the probability of classifying a frame correctly to be real or fake. We aim to estimate the probability p_V of evaluating accurately a video using n_{Samples} trials at frame level. We need the number of successful trials, i.e. the number of correctly classified frames, denoted as X , to be lower bounded by $n_{\text{Threshold}}$, which means that the number of failure trials ($n_{\text{Samples}} - X$) should be no more than $n_{\text{Samples}} - n_{\text{Threshold}}$. We can determine the value of p_V using the following formula:

$$\begin{aligned} p_V &= P(X \geq n_{\text{Threshold}}) \\ &= P(\text{number of failure cases} \leq n_{\text{Samples}} - n_{\text{Threshold}}). \end{aligned}$$

Hence, the value of p_V can be calculated using the binomial cumulative distribution function. We determine the probability of observing less than or equal to $(n_{\text{Samples}} - n_{\text{Threshold}})$ incorrectly classified frames in n_{Samples} trials, with the probability $(1 - p)$ of wrongly classified a frame on a single trial.

Figure 13 visualizes the probability p_V of successfully checking a video with different values of the probability p to successfully classify a frame. In this figure, the horizontal axis is n_{Samples} , the vertical axis is p_V , and we consider p from 0.5 to 0.9 with the step of 0.1.

We notice that for $p \geq 0.8$, the value of p_V is greater than or equal to 0.9 for $n_{\text{Samples}} > 8$, and then converges toward 1 after 20 trials. Furthermore, for $p \geq 0.9$, p_V quickly increases to 1 after 10 trials. This observation encourages us to follow the idea that we can use a frame-level classifier with high accuracy (for example, 0.9), not necessarily a perfect accuracy of up to 0.99, and use the majority voting scheme to achieve a highly reliable video-level prediction.

Now, we study the distribution of number of successful frame-level votes to demonstrate the successful chance for our proposed method to accurately classify a video to be real or fake. Using our two-stream approach, we find our

best model for $\mathcal{D}_0[50]$ with the accuracy of 0.9371. This model, together with the best model for $\mathcal{D}_0[10]$, mentioned in Table 4, is used to illustrate the potential of our method to handle video classification with majority voting scheme, as shown in Fig. 14.

In Fig. 14, we show the distribution of the number of successful votes at the frame level corresponding to $\mathcal{D}_0[10]$ and $\mathcal{D}_0[50]$, respectively. In case (a), with all bins from 6 to 10, our video-level prediction is correct. In case (b), with all bins from 26 to 50, our method can give the accurate result at the video level. For our best model in $\mathcal{D}_0[10]$, the frame-level accuracy is 0.9326, and the video-level accuracy is 0.9357, i.e. predict successfully for 655/700 videos. With our best model in $\mathcal{D}_0[50]$, the frame-level accuracy is 0.9371, and the video-level accuracy is 0.9486.

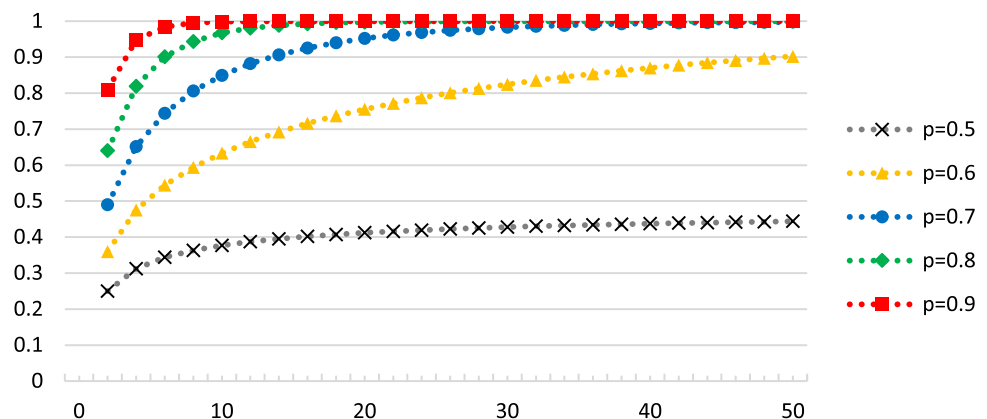
In Table 5, we show both frame- and video-level accuracy on $\mathcal{D}_0$. In this table, we also show the number of video with perfect vote results—in other words, all frames can be predicted correctly. This number is 533 and 506 for our two best models following our two-stream method in $\mathcal{D}_0[10]$ and $\mathcal{D}_0[50]$, respectively. We consider a video with a nearly perfect vote result is the video with at least 90% votes to be correct. The numbers of videos having nearly perfect vote results for the two cases are 595 and 605.

Conclusion

To make transactions, such as deposits to or withdrawals from bank accounts, customers are commonly required to go directly to the counter for authentication in the traditional KYC processes. With eKYC technology, we only need to verify online biometrics such as face scan, fingerprint, voice, or a combination of these factors on Internet/mobile Banking applications to complete identification procedures.

As face authentication is among the expected trends for online transactions and services, including eKYC, it can become a potential target for face spoofing, especially deep-fake images/videos. Therefore, in this paper, we introduce

Fig. 13 Probability to successfully check a video with different values of the probability to successfully classify a frame



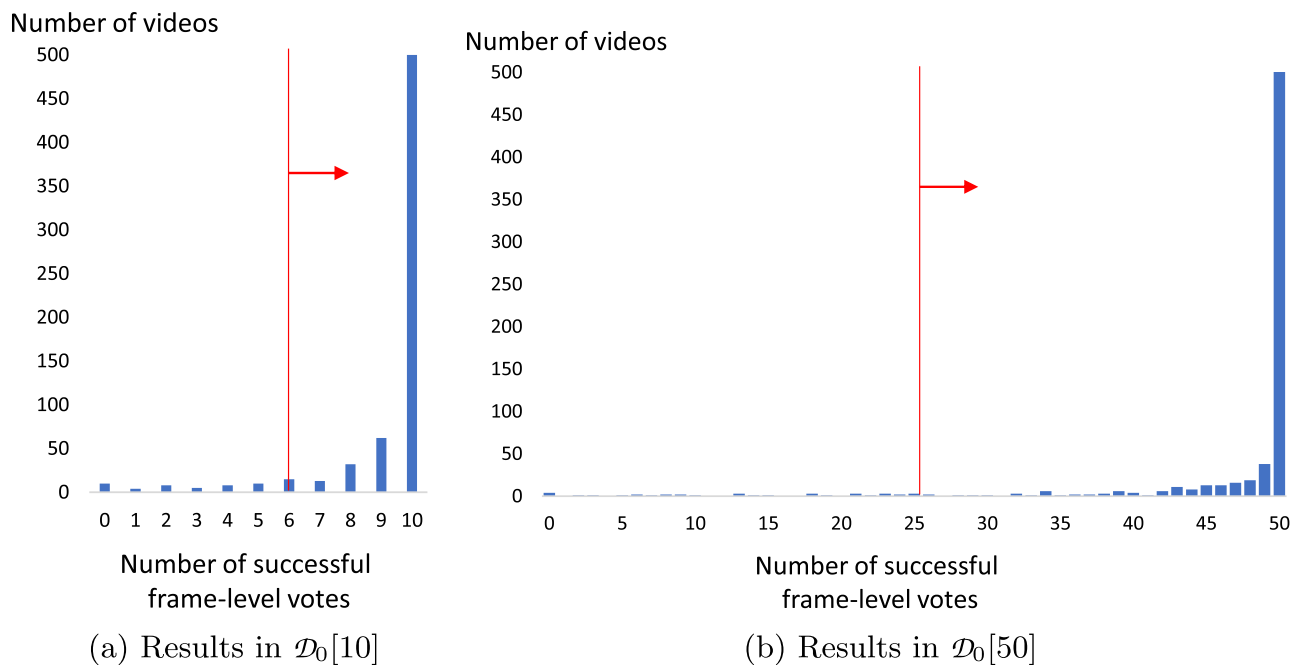


Fig. 14 Distribution of the number of successful frame-level votes

Table 5 Frame and video-level accuracy and vote status

Method	Dataset	Frame Accuracy	Video Accuracy	Correct votes
Two-stream _E	$\mathcal{D}_0[10]$	0.9326	0.9357	10 votes: 533/700 videos 9-10 votes: 595/700 videos
Two-stream _E	$\mathcal{D}_0[50]$	0.9371	0.9486	50 votes: 506/700 videos 45-50 votes: 605/700 videos

a potential threat to eKYC solutions using face swapping to synthesize/generate in real time a sequence of face images in response to facial gesture challenges from eKYC systems. We first presented our idea of this attack in Ref. [14]. Then, we propose our novel method for deepfake video detection with a two-stream network in which one branch is for a face photo, and the other is for motion-encoded input. We also propose representing the motion information of a face sequence in an average picture, which we call a motion-encoded image. This simple encoding technique helps us simplify the input of motion in a face sequence to the second branch of our two-stream network. We intend to utilize both the visual appearance and motion of a face in a photo sequence, usually captured in live mode from a smart device during the processes of an eKYC application.

Experiments on the FaceForensics⁺⁺ dataset show that using only a small number of frames in a video or a photo sequence, our method can achieve high accuracy for frame-level prediction (see the sections “[Trade-off between Number of Examined Frames and Deepfake Detection Accuracy](#)” and “[Two-Stream Network for Deepfake Detection](#)”). Using a majority voting scheme, our solution can predict correctly a video to be fake or real with high reliability, i.e. video accuracy of 0.9357 with 10 examined frames, and 0.9486 with 50 examined frames. We then propose augmenting eKYC systems’ security with Deepfake detection modules.

When we deploy our solution for actual application, we should carefully conduct more experiments to finetune and choose the optimum settings to ensure both accuracy and efficiency. However, the current promising results can be the initial steps for our future improvement and deployment. Currently, we focus our effort on improving our deepfake detection solution by setting attention to important facial blending regions and the temporal consistency of photos. We also develop modules that can be flexibly integrated into the existing eKYC systems to strengthen their security.

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Declarations

Conflict of interest All authors declare that they have no conflicts of interest.

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